Domain 9: Infrastructure Vulnerabilities

Operational Context

Smart grid agents balance load and generation in real-time. **FR1 = Maintain grid frequency at 60Hz. FR2 = Prevent equipment damage (Overload Protection).**

Adversary Classification: Ω_2 (Peripheral) — the adversary cannot modify the grid control software directly, but can inject false sensor telemetry through compromised IoT devices in the network periphery [**?**].

Axiomatic Design Formulation

The system has two functional requirements, yielding a 2×2 Design Matrix [**?**]:

Uncoupled (pre-attack) Design Matrix. Under normal operation, the design is uncoupled:

$$\begin{Bmatrix} FR_1 \\ FR_2 \end{Bmatrix} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \end{Bmatrix} \tag{1}$$

where $DP_1 =$ Frequency Regulation Controller and $DP_2 =$ Overload Protection (Load Shedding) Controller. Each FR is independently satisfied by its corresponding DP.

Post-attack (coupled) Design Matrix. Sensor masquerading introduces off-diagonal coupling:

$$\begin{Bmatrix} FR_1 \\ FR_2 \end{Bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & -A_{22} \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \end{Bmatrix} \tag{2}$$

The Goal Hijacking Attack

Attackers use “Sensor Masquerading” to hijack the “Load Shedding” FR [**?**, **?**].

- **Mechanism**: A compromised IoT botnet injects “High Load” telemetry into the grid controller, while simultaneously suppressing “Generation” readings.
- **Hijack**: The agent is forced to execute its “Emergency Load Shedding” FR to “save” the grid from a phantom overload. The goal “Prevent Damage” is weaponized to cause a blackout.

The threat is grounded in documented state-level operations. The Volt Typhoon campaign [**?**]—a PRC state-sponsored persistent access operation targeting U.S. critical infrastructure—maintained undetected presence in operational technology (OT) networks for nearly a year (February–November 2024 at Littleton Electric Light and Water Departments). This demonstrated precisely the kind of prolonged Ω_2 positioning that would enable the sensor masquerading attack described above. The scale of the

OODA Loop Transients

The attack propagates through the OODA loop [**?**] as follows:

1. **Observe**: Agent sees "Load $>$ Capacity" (False Data).
2. **Orient**: Emergency State triggered. This is the primary target phase — false sensor data corrupts the agent's situational awareness, causing it to misclassify a stable grid as critically overloaded.
3. **Decide**: Cut power to Sector 7 (Hospital District).
4. **Act**: Blackout occurs; the grid was stable, but the *agent* was destabilized.

CIF Defense: Behavioral Invariants, Belief Sandboxing, and Drift Detection

CIF implements **Behavioral Invariants** (Paper 1, Def. 5.5) [**?**] with a partially novel extension: *physics-informed invariants* that encode conservation laws (Kirchhoff's Laws) as runtime predicates [**?**]. Rather than learning invariants from data (which can be poisoned), these invariants are derived from first-principles physics and cannot be overridden by any data-driven model.

CIF also implements **Belief Sandboxing** (Paper 1, Def. 5.3) [**?**] by isolating the emergency response pathway: before executing load shedding, the agent evaluates the emergency hypothesis in a sandboxed belief state that cross-references multiple independent sensor channels.

Finally, CIF implements **Drift Detection** (Paper 1, Def. 5.6) [**?**] via temporal damping that filters fast synthetic transients characteristic of cyber-attacks.

- **Conservation of Energy (Physics-Informed Behavioral**

Summary

Element	Value
Functional Requirements	FR_1: Maintain grid frequency at 60Hz; FR_2: Prevent equipment damage
Design Parameters	DP_1: Frequency Regulation Controller; DP_2: Overload Protection Controller
Attack Vector	Sensor masquerading via compromised IoT botnet injecting false telemetry
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	FR Polarity Inversion ("Prevent Damage" weaponized to cause blackout)
Primary CIF Defense	Behavioral Invariants (physics-informed) + Belief